AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Artificial Intelligence in the Contemporary World

Employee Payroll data cleaning and financial analysis

[Excel / Power BI]

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

by:

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3. Problem Statement

1. Payroll Errors, Employee Trust and Attrition

One major business problem is payroll errors that impact employee trust and retention. In many organizations, issues such as incorrect salary calculations, wrong deductions, delayed payments, and lack of transparency in payslips are quite common. These errors lead to dissatisfaction, increased complaints, and reduced motivation among employees, which can ultimately result in higher attrition. Since payroll is directly linked to employee satisfaction, even small mistakes can have a significant impact. This project addresses the problem by ensuring payroll data is clean, accurate, and properly validated, thereby improving trust and reliability in the system.

2. Financial Loss, Overpayment and Fraud Risk

Another critical business problem is financial loss caused by overpayments, duplicate records, and fraud risks such as ghost employees. In large organizations, even small errors in payroll can accumulate into significant financial leakage over time. Additionally, the presence of inactive or fake employees in the system can lead to unauthorized salary payments. These issues directly affect the company's profitability and financial control. This project helps mitigate these risks by identifying duplicates, validating records, and ensuring that payroll data is accurate and reliable.

3. Poor Financial Planning and Lack of Insights

Organizations also face challenges in financial planning due to unstructured and inaccurate payroll data. Without proper analysis, management cannot clearly understand total payroll costs, department-wise expenses, or salary trends. This lack of insights leads to poor budgeting, inefficient decision-making, and difficulty in controlling costs. Payroll is a major expense for any company, so the absence of clear insights can impact overall financial strategy. This project addresses this issue by performing structured analysis and generating key insights that support better planning and decision-making.

4. Manual Processes, Inefficiency and Scalability Issues

Many organizations still rely on manual payroll processes, which are time-consuming and prone to human error. HR and finance teams often spend significant time verifying data, correcting mistakes, and processing salaries. As the organization grows, the volume of data increases, making these manual processes inefficient and difficult to scale. This reduces productivity and increases operational costs. This project improves efficiency by applying systematic data cleaning and structured analysis, reducing manual effort and making the process more reliable and scalable.

5. Compliance Risk, Data Inconsistency and Security Issues

Payroll systems also face compliance and data management challenges. Errors in salary calculations, tax deductions, or statutory payments can lead to legal and regulatory issues for the organization. At the same time, inconsistent data formats and poor data handling practices increase the risk of data security and confidentiality breaches. Since payroll data is highly sensitive, any mismanagement can have serious consequences. This project addresses these challenges by ensuring data consistency, validation, and structured handling, which helps reduce compliance risks and improves data security.

4. Objectives

- To collect and explore a large-scale payroll dataset from Kaggle and understand its structure, including fields such as Employee ID, Department, Gross Salary, Bonus, Deductions, and Net Salary.
- To clean and preprocess the payroll data by removing duplicate records, handling missing values, correcting data formats, and standardizing department names and salary fields to ensure accuracy.
- To validate payroll calculations by ensuring that Net Salary is correctly calculated as Gross Salary minus Deductions and identifying any inconsistencies in the dataset.
- To perform payroll analysis using Excel formulas such as SUM, AVERAGE, MAX, MIN, COUNTIF, and basic Pivot Tables to derive insights like total payroll cost, average salary, and department-wise salary distribution.
- To generate a structured analytical report that clearly presents key insights related to payroll costs, salary trends, and potential inconsistencies for better decision-making.
- To integrate AI tools such as ChatGPT or Microsoft Copilot to assist in formula validation, error detection, and generating structured summaries of the analysis.
- To document the complete workflow, including data cleaning steps, analysis process, and AI usage, and prepare the final project submission along with required files and demonstration.

5. Methodology / Implementation Plan

Provide a simple overview of how the project will be carried out. The following five-phase workflow applies to all projects under this fellowship program.

#	Phase	Description
1	**Data Collection & Exploration**	The payroll dataset was obtained from Kaggle in Excel format. The dataset includes fields such as Employee ID, Department, Gross Salary, Bonus, Deductions, and Net Salary. Initial exploration was performed to

		understand the structure, data types, total number of records, and identify common data quality issues such as missing values, duplicate entries, and inconsistent formats.
2	**Data Cleaning & Preprocessing**	Data cleaning was carried out by removing duplicate records using Excel's "Remove Duplicates" feature. Missing values were handled using functions like IF and IFERROR. Data types were corrected, and inconsistencies in department names and salary formats were standardized. Salary calculations were also validated to ensure that Net Salary equals Gross Salary minus Deductions.
3	**Analysis & AI Integration**	Payroll analysis was performed using Excel formulas such as SUM, AVERAGE, MAX, MIN, COUNTIF, and SUMIF to calculate key metrics like total payroll cost, average salary, and department-wise distribution. Basic Pivot Tables were used for summarization. AI tools such as ChatGPT or Microsoft Copilot were used to validate formulas, identify errors, and generate structured summaries of insights.
4	**Visualization / Output**	A structured analytical report was prepared to present key findings and insights from the payroll data. The focus was on clearly communicating salary trends, cost distribution, and identified inconsistencies. The output emphasizes accuracy and clarity rather than complex visual dashboards.
5	**Documentation & Submission**	The complete project process, including data cleaning steps, analysis methods, and AI usage, was documented. All relevant files were prepared for submission, along with a clear explanation of the workflow and results. A demonstration video will be recorded to present the methodology and key outcomes.(if required). And all of the documents will be upload in github.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i3 or higher, minimum 4 GB RAM (8 GB recommended)
- Operating System: Windows 10 / 11
- Microsoft Excel (Microsoft 365 / Excel 2019 or later)
- Power BI Desktop — latest version (free download from Microsoft) — where applicable
- Microsoft Copilot — via Microsoft 365 subscription — where applicable
- AI Chatbot (ChatGPT / Google Gemini / Microsoft Copilot Web) — for prompt-based projects
- Internet connection — required for Copilot and AI Chatbot access
- Web browser — Google Chrome or Microsoft Edge (latest version)

B. Dataset

- The dataset was obtained from the kaggle
- Source link if publicly available - https://www.kaggle.com/datasets/dsfelix/employees-payroll-in-los-angeles

C. Project Timeline

A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*
2	Data cleaning, preprocessing, and validation / Prompt design	*Week 2*
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	*Week 2 – 3*
4	Dashboard / output build, testing, and review	*Week 4/5*
5	Report writing, GitHub upload, and video recording and submission	*Week 5/6*

7. Expected Outcomes / Conclusion

A clean and validated payroll dataset that reduces errors in salary calculations, helping improve accuracy and build employee trust in payroll processing.

Identification and removal of duplicate or incorrect records, reducing the risk of overpayments, financial loss, and potential fraud such as inactive or duplicate employee entries.

A structured analysis of payroll data providing clear insights such as total payroll cost, department-wise expenses, and salary distribution, supporting better financial planning and decision-making.

Improved efficiency in payroll data handling by reducing manual errors and organizing the data in a way that is easier to manage and scalable for larger datasets.

Consistent and standardized data format that helps reduce compliance risks and ensures better handling of sensitive payroll information.

A clear and well-documented analytical report that can be used by HR and finance teams to understand payroll trends, identify issues, and take corrective actions.

Practical demonstration of how Excel and AI tools can be used together to improve accuracy, reduce effort, and enhance overall payroll analysis.

8. References

- Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available at: https://www.edunetfoundation.org
- LinkedIn Learning. AI and Data Analytics Courses. 2026. Available at: https://www.linkedin.com/learning
- Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel
- Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/
- Microsoft Corporation. Microsoft Copilot. Available at: https://copilot.microsoft.com
- Kaggle Inc. Open Datasets. Available at: https://www.kaggle.com/datasets